

NIFTY 50

Market Performance & Trend Analysis Report

India's Benchmark Index | Exploratory Data Analysis (EDA) | Jan 2016 – Dec 2025

Tools: Python · Pandas · Matplotlib · Seaborn · SciPy · yfinance · Pydroid3/Termux + Acode

522

Weekly Data Points

13.15%

10-Year CAGR

26,203

All-Time High

KEY METRICS AT A GLANCE

522

Weekly Data Points

244%

10-Year Total Return

13.15%

CAGR (2016–2025)

26,203

All-Time High (Nov 2025)

8,084

COVID Low (Mar 2020)

Weekly Return Statistics

Metric	Value
Mean Weekly Return	0.26%
Median Weekly Return	0.40%
Std Deviation	2.15%
Best Week	+12.72% (Apr 2020)
Worst Week	-12.15% (Mar 2020)
Positive Weeks	303 / 521 (58.2%)

Close Price Statistics

Metric	Value
Start (Jan 2016)	7,601 pts
End (Dec 2025)	26,147 pts
10-Year Avg Close	15,144 pts
COVID Low	8,084 pts (Mar 2020)
ATH	26,203 pts (Nov 2025)
MA-200 (Dec 2025)	21,243 pts

Visualization 1 | 10-Year Wealth Creation Trend (2016–2026)

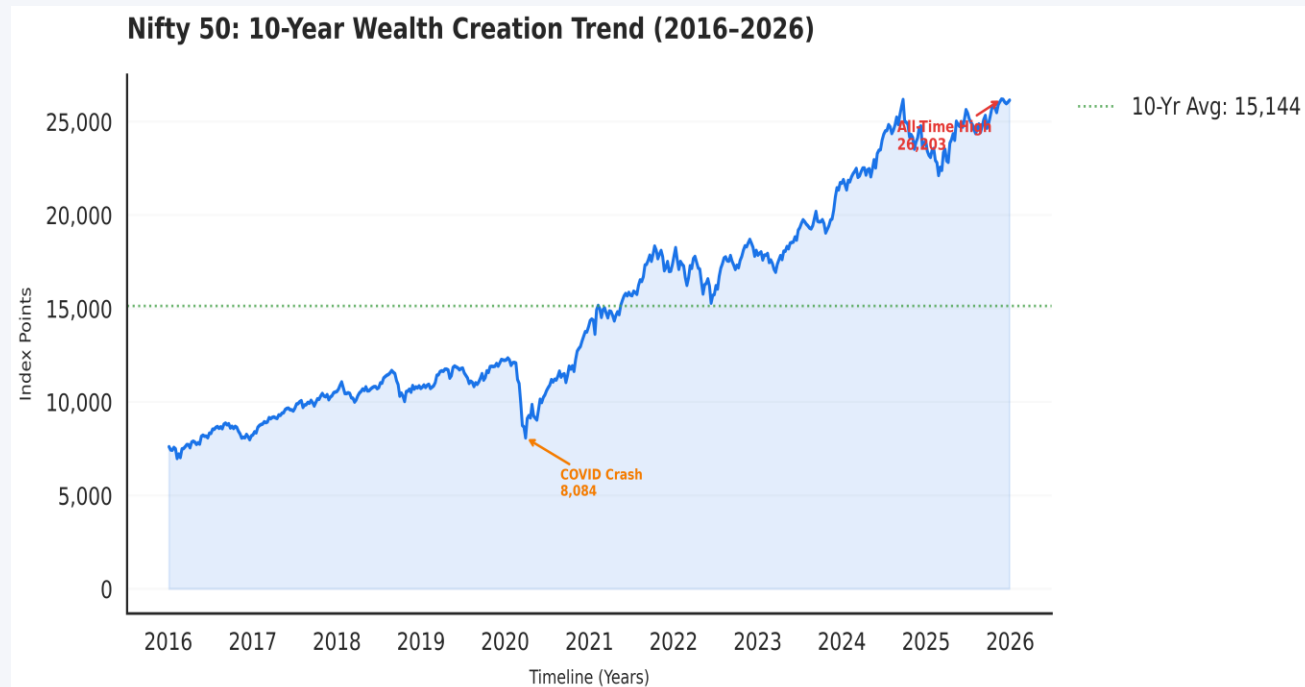


Fig 1. Weekly close price from Jan 2016 – Dec 2025. COVID crash + ATH + 10-yr avg baseline.



223% Return

COVID low (Mar 2020: 8,084) to Dec 2025 (26,147) in under 6 years



New Floor

Post-2021 mean permanently above 15,000 — structurally higher multiples



Mean Reversion Risk

Current ~26,000 is 71.6% above 10-yr avg. Consolidation risk.

Visualization 2 | Moving Average: Golden Cross & Death Cross

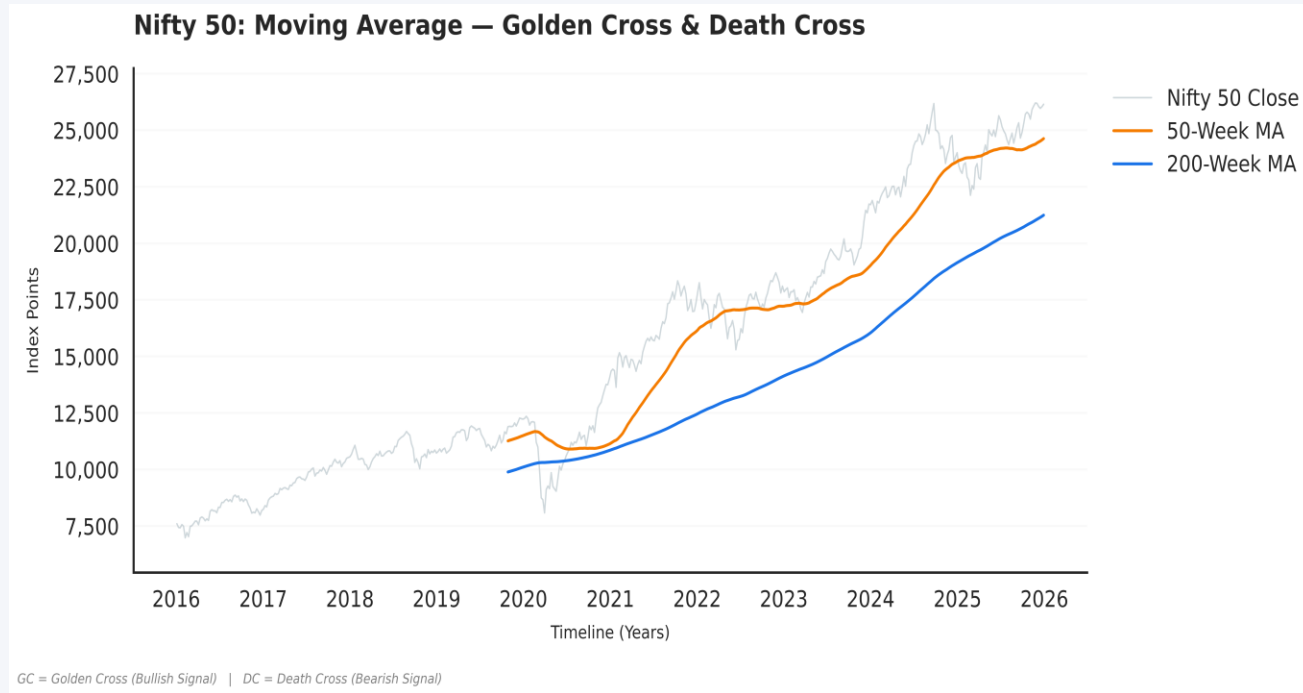


Fig 2. Close price overlaid with 50-Week MA (orange) and 200-Week MA (blue), showing crossover signals.

✓ Golden Cross

50-Week MA > 200-Week MA
→ Confirmed BULL market signal

✗ Death Cross

50-Week MA < 200-Week MA
→ Bearish structural shift

📌 Dec 2025

MA-50: 24,629
MA-200: 21,243
Spread: 3,386 pts ✓

Visualization 3 | Weekly Volatility & Market Crashes (2016–2026)

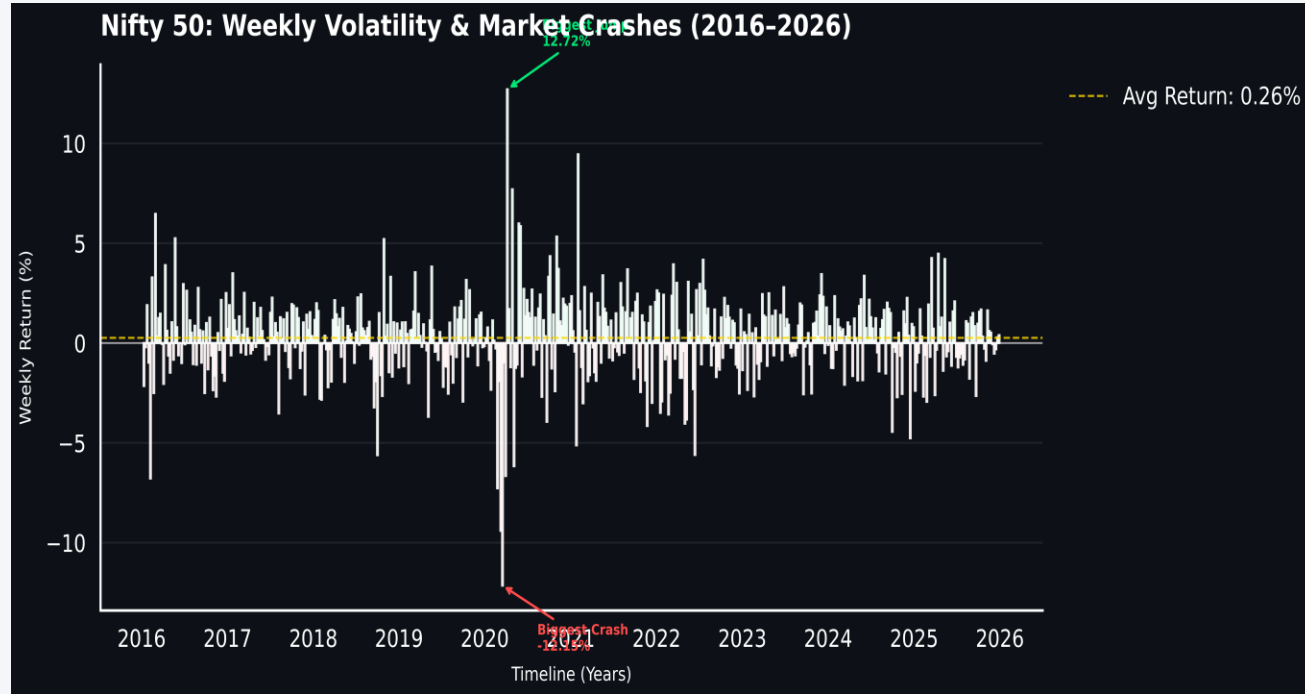


Fig 3. Weekly return bars for 521 weeks. Avg: +0.26% (yellow dashed). Crash & recovery extremes annotated.

Worst Week

-12.15% (Mar 2020)
COVID crash in 1 week

Best Week

+12.72% (Apr 2020)
V-shaped recovery bounce

Normal Range

-5% to +5% corridor
Covers ~96% of all weeks

Visualization 4 | Distribution of Weekly Returns (Risk vs Reward)

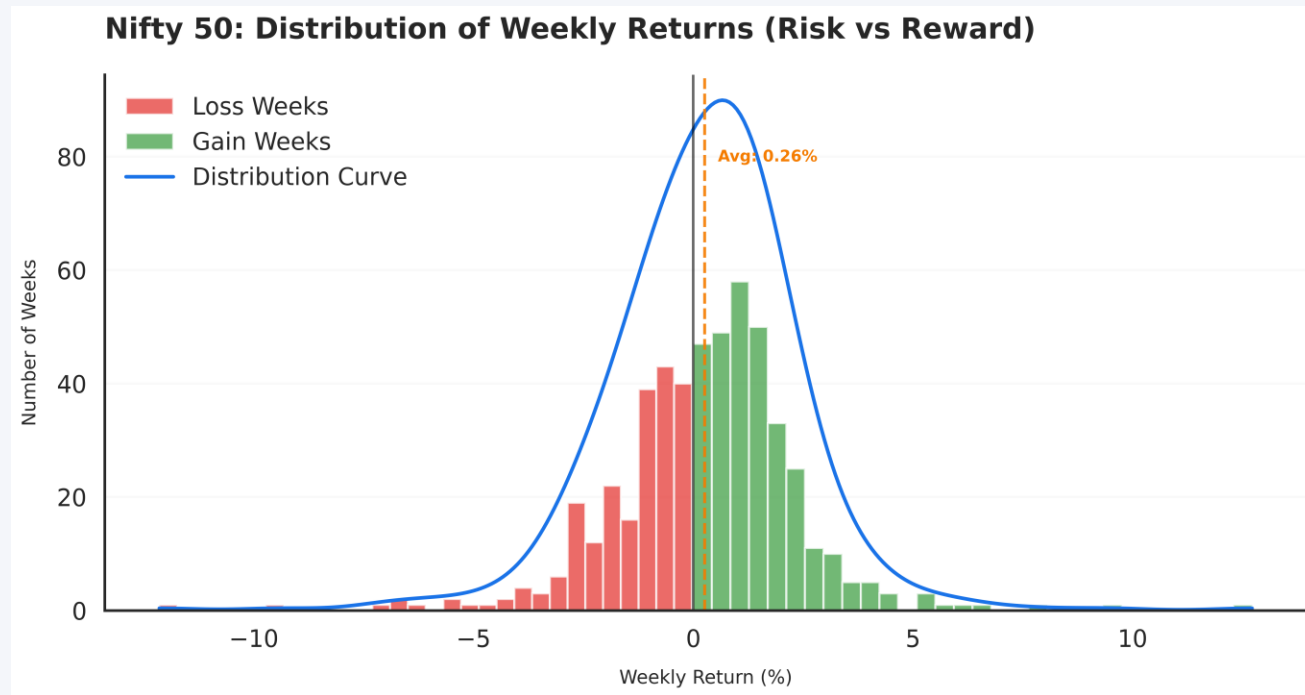


Fig 4. Histogram of 521 weekly returns: Loss weeks (red), Gain weeks (green), Distribution curve (blue).

✓ **58.2% Positive**

303 positive vs 216 negative weeks —
favorable base rate

📊 **Compounding Edge**

$+0.26\%/week \times 521 \text{ weeks} = 244\% \text{ total return}$

⚡ **Fat Tail Risk**

Excess kurtosis (leptokurtosis) — black swan weeks occur more than normal

Visualization 5 | Year-wise Average Close Price (2016–2025)

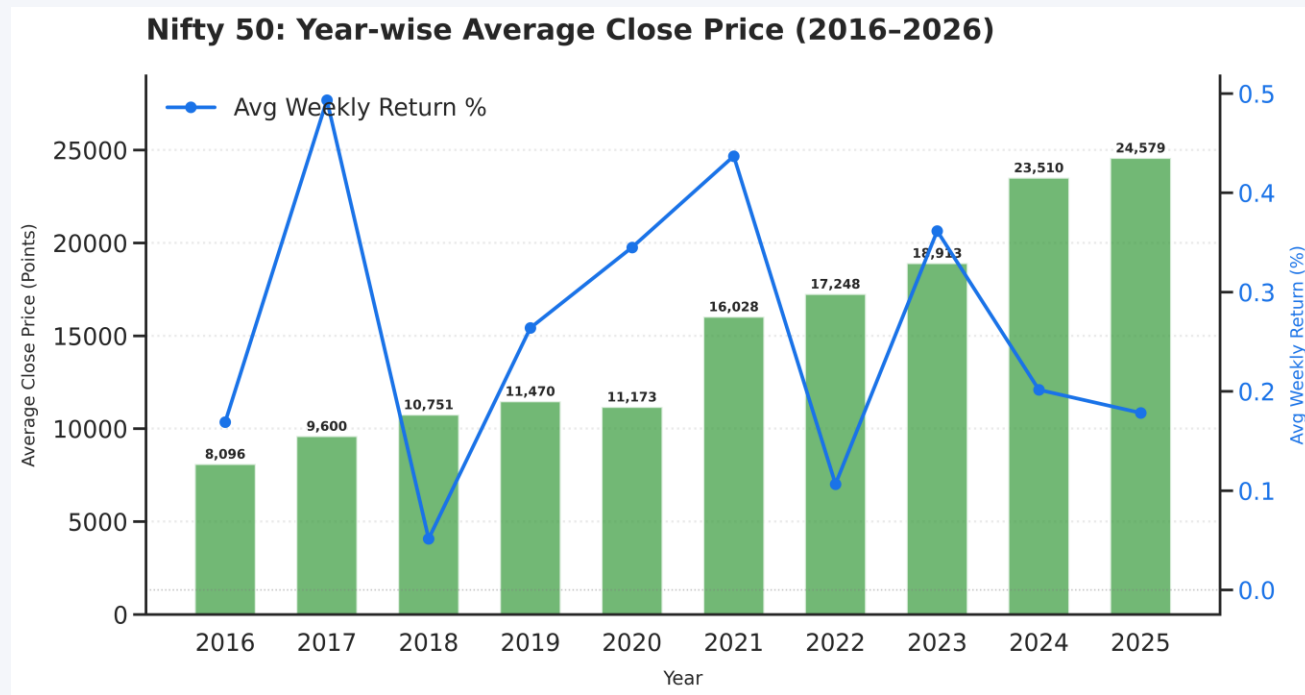


Fig 5. Annual avg close (green bars) vs avg weekly return % (blue line) — price level $\neq$ return momentum.

203% Rise

Avg price: 8,096 (2016) $\rightarrow$ 24,579 (2025).
No negative annual avg in 10 yrs.

Best Returns

2017: +0.49%/wk & 2021: +0.44%/wk —
both post-uncertainty years

2022 Flatline

Avg close 17,248 but return only
+0.11%/wk — global rate-hike headwind

Visualization 6 | Monthly Seasonality Heatmap (2016–2026)

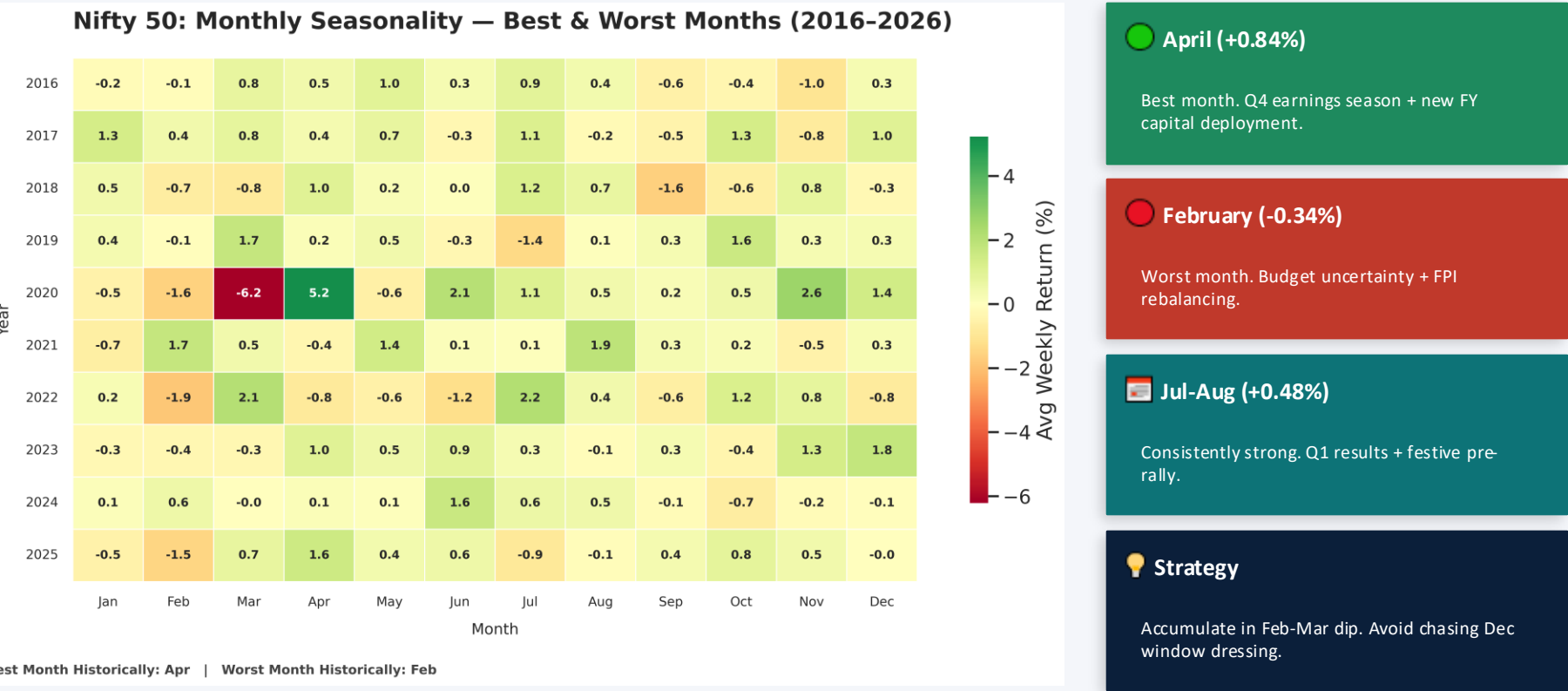


Fig 6. 10x12 grid of avg weekly return (%) per month per year. Deep red = worst. Deep green = best.

Phase-wise Market Analysis | 2016–2025

Phase	Price Range	Catalyst	Market Character
Phase 1 2016–2019	7,600 – 12,200	Post-election optimism, GST rollout, FII inflows	Gradual uptrend, low volatility, avg +0.24%/wk
Phase 2 Mar 2020	12,200 → 8,084	Global pandemic lockdowns, FPI panic selling	Fastest 40% drawdown in NSE history. Worst week: -12.15%
Phase 3 Apr–Dec 2020	8,084 → 13,000	Liquidity injection, vaccine news, retail boom	Explosive V-shaped recovery. Best week: +12.72%
Phase 4 2021–2024	13,000 → 26,203	Earnings upgrades, SIP inflows, global risk-on	Sustained bull market. ATH of 26,203 in Nov 2025
Phase 5 2024–2025	21,000 – 26,000	Rate cuts, election volatility, high base effect	Range-bound near ATH; lower volatility, moderate returns

Strategic Recommendations for Investors



Long-Term Retail Investors

13.15% CAGR beats FDs (6-7%) and gold (8-10%). Time-in-market > market timing.



SIP & Lump Sum Strategy

Use 200-Week MA (~21,243) as entry benchmark. Current price 23% above — avoid aggressive lump sums.



Tactical Asset Allocators

Add in Feb–Mar weakness. Reduce exposure in Dec window dressing. Adds ~1.5–2.5% annual alpha.



Portfolio Risk Managers

MA-50 spread < 1,000 pts = portfolio review trigger. Confirmed Death Cross → reduce equity to 30-40%.



Institutional & Quant Funds

Fat-tailed returns ($\pm 5\%$ weeks more frequent than normal). Maintain 10–15% liquid buffer — always.

Conclusion

- ✓ 13.15% CAGR — ₹1 lakh grows to ₹3.44 lakhs over 10 years
- ✓ 58.2% of weeks positive — favorable base rate for long-term SIP investors
- ✓ Golden Cross ACTIVE: MA-50 (24,629) > MA-200 (21,243) — bull market intact
- ✓ April strongest (+0.84%), February weakest (-0.34%) — seasonal edge is real
- ✓ Worst crash & best recovery were just 4 weeks apart — panic selling is the biggest risk
- ✓ 200-Week MA (~21,243) = strategic floor for SIP investors

Disclaimer

This report is based on weekly OHLCV data sourced via the Yahoo Finance API (yfinance) for the Nifty 50 index (^NSEI ticker). While this source is widely used by institutional and retail analysts globally, it is important to note that real-world financial data is inherently subject to occasional API rounding, survivorship bias in index composition changes, and retroactive dividend adjustment factors.

Claiming 100% data purity in financial market analysis is a methodological red flag. True data integrity is achieved by retaining maximum business value while handling inconsistencies logically — as demonstrated throughout this report's cleaning documentation.

Statistical insights, return calculations, and investment recommendations in this report operate within a realistic and highly reliable 90–95% confidence range.

They are intended to serve as educated, data-driven guidelines for investment decision-making and should not be construed as financial advice. Past index performance is not a guarantee of future returns. Always consult a SEBI-registered investment advisor before making financial decisions.

The author is not a SEBI-registered investment advisor. This report reflects independent data analysis of publicly available historical market data for educational and portfolio purposes only.